

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Cloud and Orchestrate”

Project Tile – AI RESEARCH AGENT

Domain of Project – RESEARCH

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Problem Statement -

RESEARCH AGENT

The Challenge

Researchers, students, and professionals often deal with scattered academic papers, technical articles, and experimental data spread across multiple platforms. Filtering relevant knowledge, summarizing insights, and keeping track of the latest developments is time-consuming and inefficient. The challenge is not just to collect references but to synthesize, analyze, and generate actionable research insights in real time.

The Objective

Build an agentic AI application that acts as an intelligent research companion. The solution should:

1. Fuse Academic Sources: Aggregate data from research papers, journals, preprints, and knowledge repositories, then synthesize them into clear, concise literature reviews.
2. Enable Multimodal Research Input: Accept user queries, notes, charts, or even PDFs, and convert them into structured insights, references, and summaries.
3. Create a Predictive & Agentic Layer: Highlight emerging trends, detect citation gaps, and predict promising directions for future research in a given domain.
4. Visualize Research Insights: Provide an interactive dashboard with topic clusters, citation maps, sentiment/trend analysis, and concise knowledge graphs.

Tech Stack (Mandatory)

- IBM Langflow / IBM Orchestrate + IBM Models

Proposed Solution -

- ResearchX – An Agentic AI Research Companion

- Solution Overview

ResearchX is an AI-powered research assistant built using IBM Langflow / IBM Orchestrate and IBM Granite Models (watsonx.ai). It helps researchers, students, and professionals discover, analyze, summarize, and visualize research from multiple academic sources.

Instead of acting like a simple search engine, ResearchX functions as an agentic AI system that plans tasks, gathers information from multiple sources, verifies results, identifies research gaps, predicts future trends, and presents insights through an interactive dashboard..

- Problem Being Solved

- 1.Researchers spend significant time:
- 2.Searching across multiple databases.
- 3.Reading hundreds of papers.
- 4.Organizing references manually.
- 5.Identifying research gaps.
- 6.Tracking emerging technologies.
- 7.Preparing literature reviews.

- ResearchX automates these tasks using intelligent AI agents.

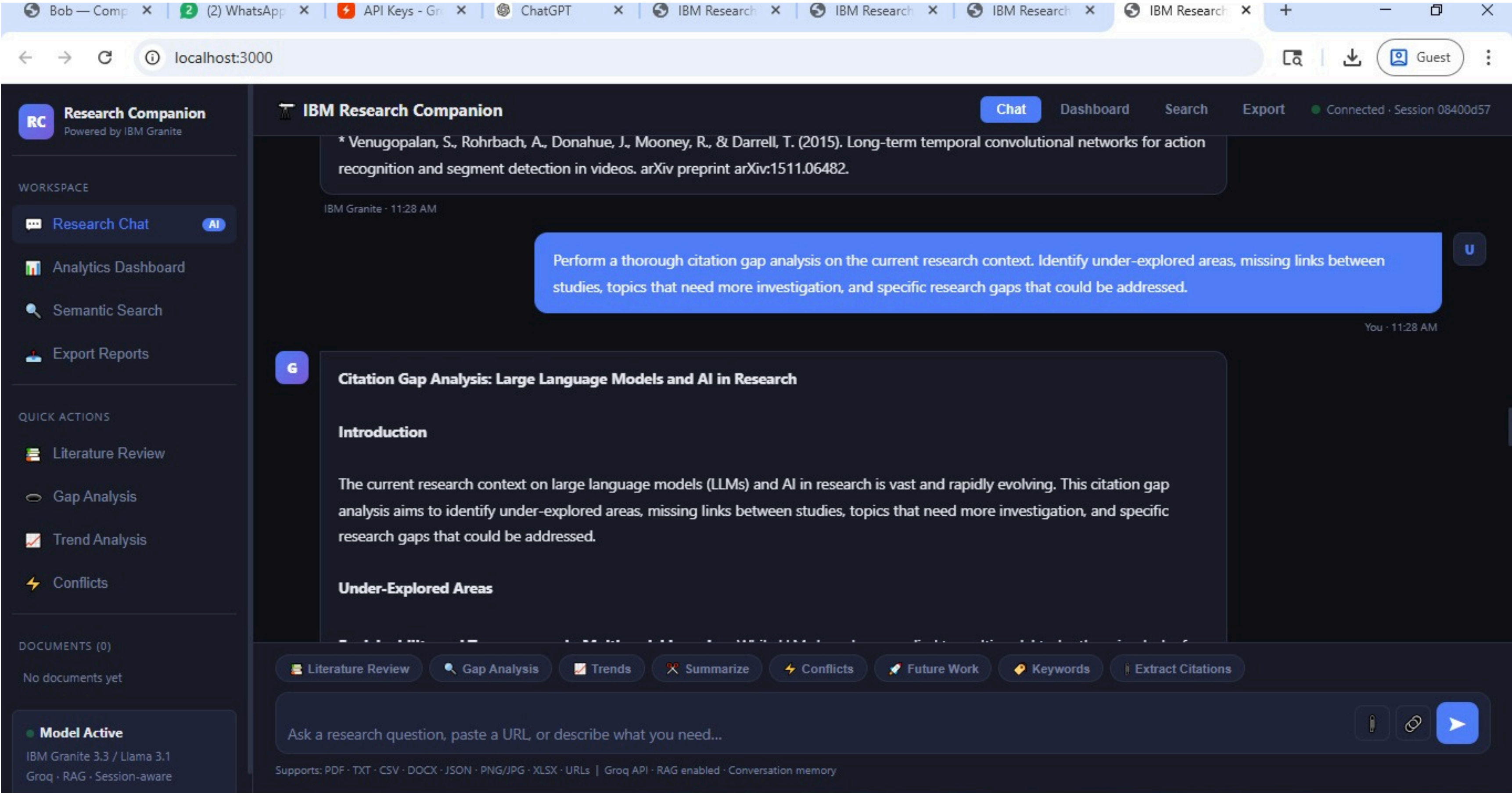
Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **IBM BOB** -an AI-powered, agentic software development partner designed to assist enterprise teams with planning, coding, testing, and modernizing legacy applications. [1, 2]

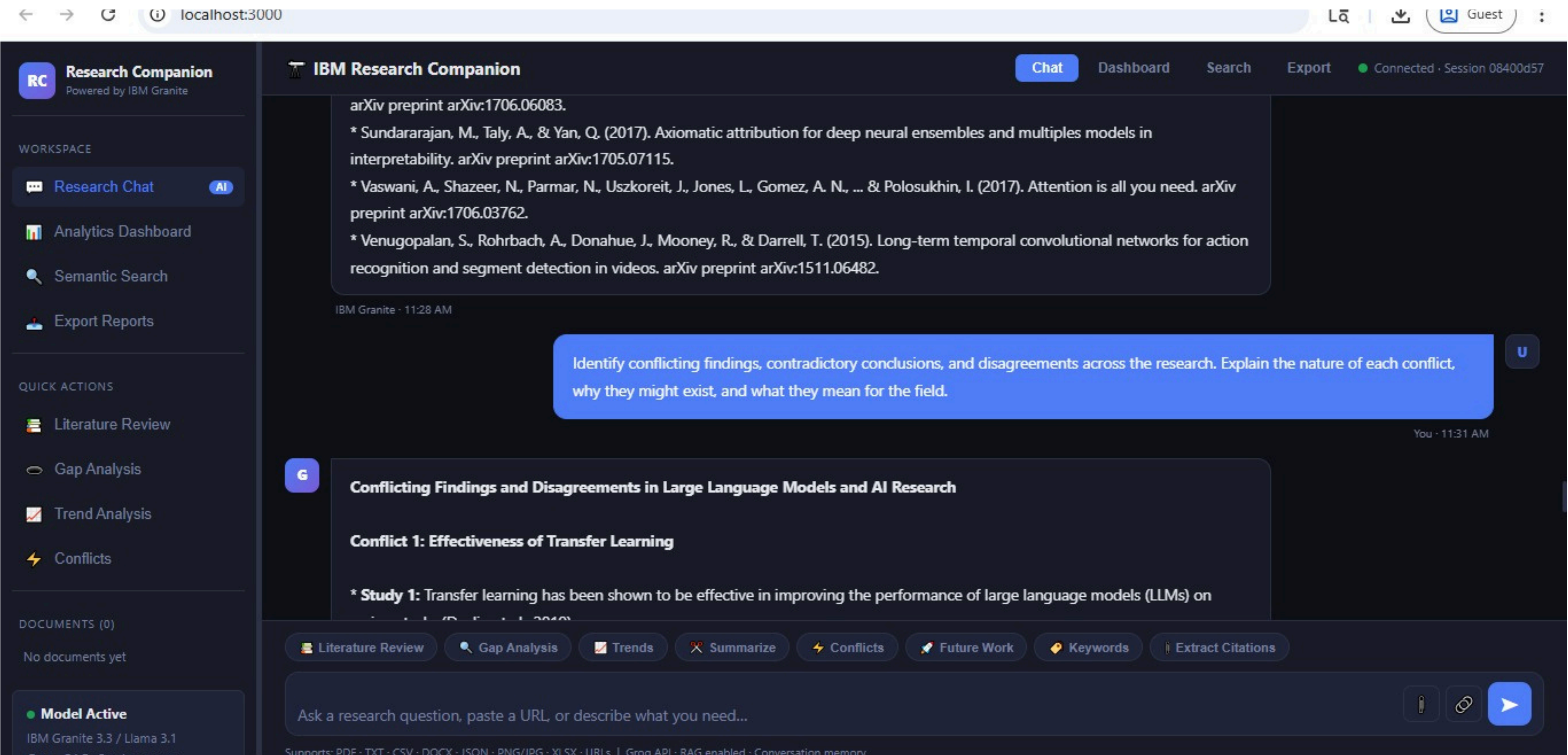
Orchestrate component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, reference booksetv (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate the information about the research papers authors etc
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations
- 4) **Chat output** -Displays AI-generated responses for research papers books etc.
- 5) **File component** - **Gives info about all the resaerches related to your searched file.**

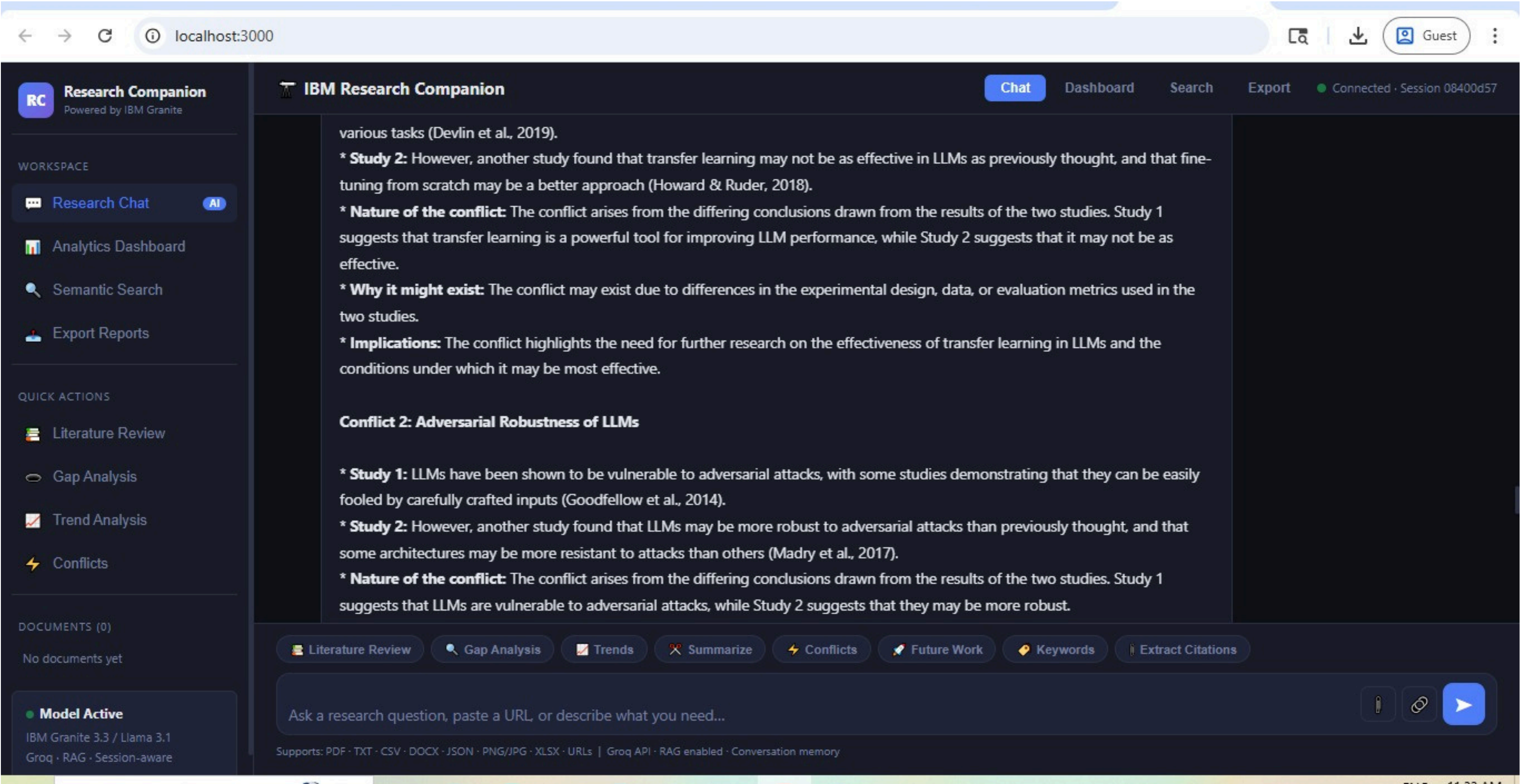
Agent Instructions Screenshot-



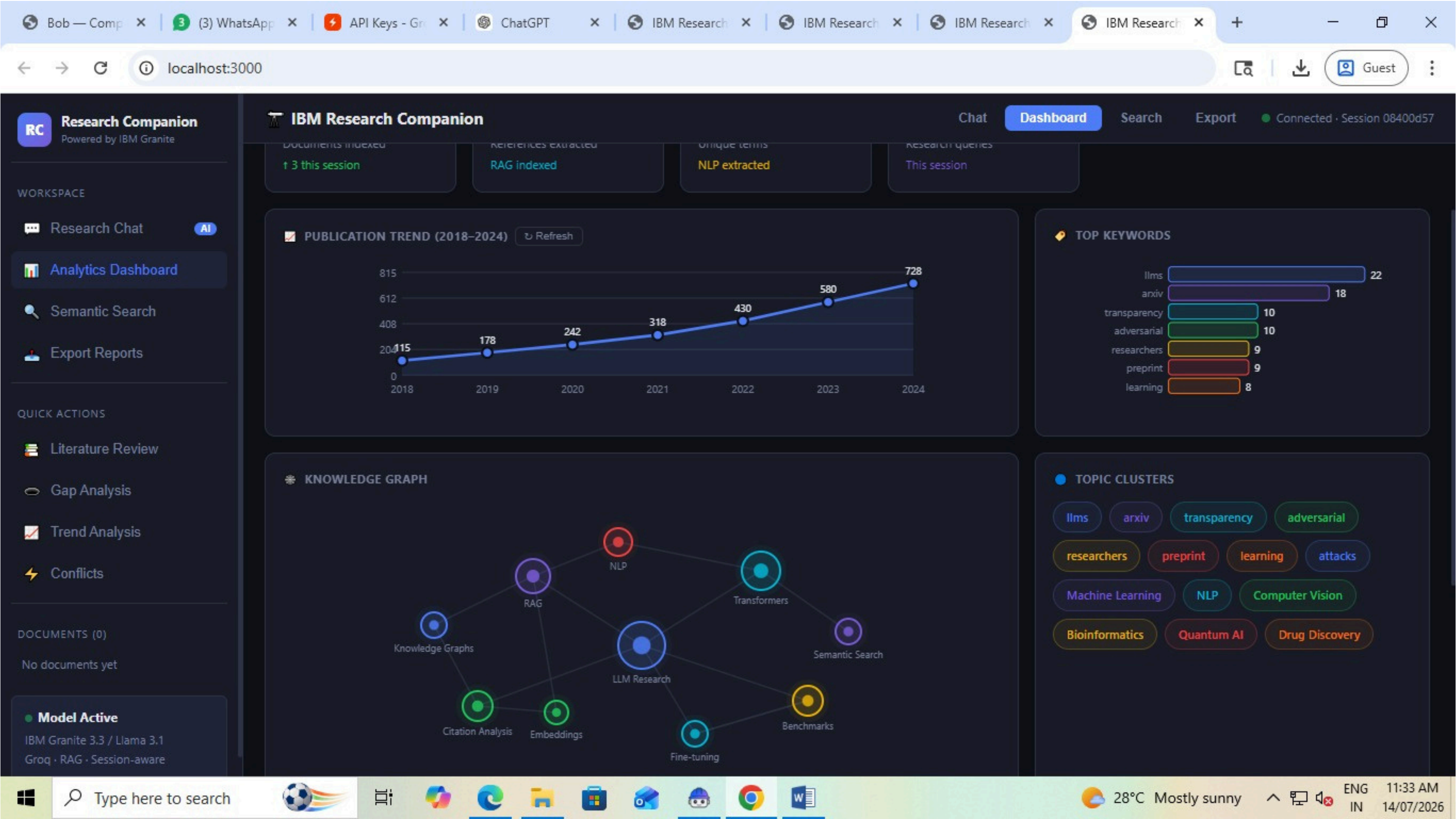
Document Uploaded Screenshot -



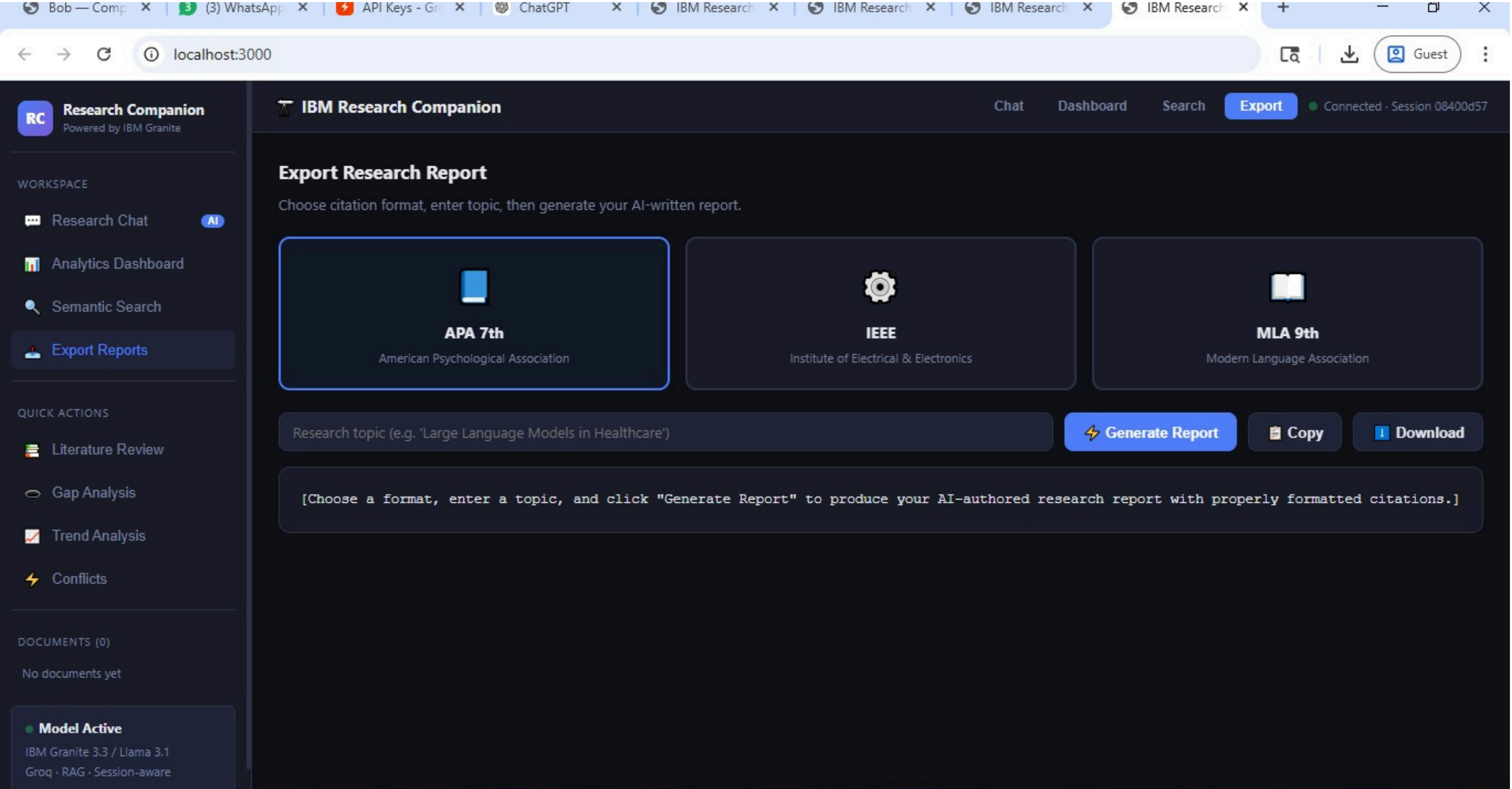
Prompt with output Screenshot:



Other Applicable Screenshots:



Orchestrate architecture blueprint



Role of Agentic AI in Research Agent:

- > ****Agentic AI enables the Research Agent to function as an intelligent, autonomous research assistant rather than a simple information retrieval tool. Using IBM watsonx.ai and IBM Granite models, it coordinates multiple specialized agents to help researchers discover, analyze, and synthesize knowledge from diverse sources.**
- > ****Each agent performs a specific role**—such as retrieving information from research papers, journals, and trusted websites; summarizing findings; comparing multiple sources; identifying knowledge gaps; extracting key insights; and generating well-structured research reports—while working collaboratively. This multi-agent approach ensures comprehensive research, efficient knowledge discovering.**

NOVELTY AND UNIQUENESS

Novelty and Uniqueness of the Research Agent

Multi-Agent Intelligence – Multiple specialized AI agents collaborate to retrieve, analyze, summarize, validate, and organize research efficiently.

RAG-Based Knowledge Retrieval – Combines Retrieval-Augmented Generation (RAG) with AI to access real-time, reliable information from research papers, journals, and trusted sources.

Personalized Research Assistance – Delivers tailored literature recommendations, summaries, and research insights based on the user's domain and objectives.

Intelligent Gap Analysis – Identifies research gaps, compares findings across sources, and suggests future research directions.

Multimodal Research Support – Processes text, PDFs, images, charts, and voice inputs for comprehensive knowledge extraction.

AI-Powered Research Automation – Automates literature review, summarization, citation extraction, and report generation to save time and improve accuracy.

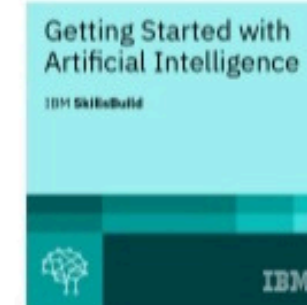
Scalable & Enterprise-Ready – Built on IBM watsonx.ai, IBM Granite models, and LangFlow for secure, scalable, and governed AI workflows.

Github link:

<https://github.com/samarthhatte07-sys/RESEARCH-AGENT>

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Learning hours: 30 mins

Thank You!

Thank you for your time and interest.